

RCIM Model-Bank Reproduction Exact-Paper Open-Cell Repair Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the exact-paper open-cell repair campaign prepared in:

- `doc/reports/campaign_plans/track_1/exact_paper/2026-04-13-21-20-53_track1_exact_paper_open_cell_repair_campaign_plan_report.md`

The campaign executed 6 exact-paper validation runs through the dedicated launcher:

- completed runs: 6
- failed runs: 0
- execution window: 2026-04-13 22:07:17+02:00 to 2026-04-13 22:09:59+02:00
- campaign artifact root:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/` This campaign was designed under the new RCIM Model-Bank Reproduction paradigm:
- the primary objective is exact-paper table closure across Tables 3-6 ;
- the campaign winner remains a required bookkeeping artifact;
- however, the winner does not define campaign success by itself.

Objective And Outcome

The campaign had three concrete goals

1. refresh the exact-paper baseline under the new cell-closure-first framing;
2. test whether paper-aligned family subsets can close any currently open cells in Tables 3-5 ;
3. reduce the number of harmonics still marked `not_yet_matched_tables_3_6` . Outcome:

- all 6 runs completed successfully;
- all 6 runs exported ONNX artifacts with 0 failed targets;
- no run closed any new numeric paper-target cell in Tables 3-5 ;
- the best campaign runs reduced Table 6 open harmonics from 3 to 2 ;
- specifically, harmonic 240 moved from `not_yet_matched_tables_3_6` to `partially_matched_tables_3_6` ;
- no harmonic became fully closed.

The campaign therefore improved harmonic-family alignment slightly, but it did not close the real numeric gaps that still keep RCIM Model-Bank Reproduction open.

Ranking Policy

This campaign cannot be ranked primarily by winner MAPE , because all runs produced the same winning-family aggregate metric. The explicit serialized campaign policy is therefore:

- primary metric: harmonic_closed_count_desc
- first tie breaker: harmonic_open_count_asc
- second tie breaker: met_paper_cell_count_desc
- third tie breaker: matched_table6_family_target_count_desc
- fourth tie breaker: regression_count_asc
- fifth tie breaker: enabled_family_count_desc
- sixth tie breaker: run_name

Interpretation:

- this policy exists only to serialize the required campaign-best artifacts;
- the real RCIM Model-Bank Reproduction interpretation remains the cell-closure matrix below.

Campaign Ranking

Ranked Completed Runs

Rank	Config	Family Scope	Harmonics Open	Harmonics Partial	Met Cells	Regressions
1	exact_open_cell_paper_family_reference	6 paper-selected families	2	8	14	0
2	exact_open_cell_phase_gap_bridge	5 phase-bridge families	2	8	14	0
3	exact_open_cell_refresh_full_bank	all 10 families	3	7	14	0
4	exact_open_cell_tree_control_bank	8 tree-heavy families	3	7	14	0
5	exact_open_cell_low_order_repair	5 low-order families	3	7	11	3
6	exact_open_cell_high_order_tree_repair	3 tree families	4	6	11	3

Config	Winner	Mean Component MAPE [%]	Export Mode	Exported	Fail.
exact_open_cell_paper_family_reference	RF	18.369	strict	120	0
exact_open_cell_phase_gap_bridge	RF	18.369	strict	100	0
exact_open_cell_refresh_full_bank	RF	18.369	continue	200	0
exact_open_cell_tree_control_bank	RF	18.369	strict	160	0
exact_open_cell_low_order_repair	RF	18.369	strict	100	0
exact_open_cell_high_order_tree_repair	RF	18.369	strict	60	0

Campaign Best Run

The explicit bookkeeping winner is

- exact_open_cell_paper_family_reference

It was selected because it achieved:

- the minimum harmonic-open count in the campaign: 2
- the maximum harmonic-partial count in the campaign: 8
- zero numeric regressions versus the canonical baseline
- the broadest paper-aligned family coverage among the tied top runs

Important interpretation:

- this run is the campaign bookkeeping winner;
- it is **not** evidence that RCIM Model-Bank Reproduction is solved;
- it is only the cleanest top-ranked run under the paper-closure-first ranking policy.

Cell-Closure Impact

Tables 3-5

There was no numeric closure progress in the campaign.

The still-open numeric cells after the best run remain:

Table	Harmonics Still Open
Table 3 amplitude RMSE	0 , 1 , 3 , 81 , 240
Table 4 phase MAE	1 , 3 , 162 , 240
Table 5 phase RMSE	1 , 3 , 39 , 162 , 240

Table 6

The campaign did produce one harmonic-status improvement:

- harmonic 240 moved from not_yet_matched_tables_3_6 to partially_matched_tables_3_6 Best achieved harmonic status after the campaign:
- fully closed harmonics: 0
- partially matched harmonics: 8
- still-open harmonics: 2 -> 0 , 1

Open Numeric Gaps After The Best Run

Cell		Paper Family	Repository Family		Gap To Paper
Table 3	A_0 RMSE	SVM	HGBM		0.000399
Table 3	A_1 RMSE	RF	HGBM		1.76e-07
Table 4	phi_1 MAE	LGBM	HGBM		4.64e-05
Table 5	phi_1 RMSE	HGBM	GBM		1.01e-05
Table 3	A_3 RMSE	HGBM	HGBM		7.42e-07
Table 4	phi_3 MAE	HGBM	GBM		0.003757
Table 5	phi_3 RMSE	HGBM	HGBM		0.005302
Table 5	phi_39 RMSE	HGBM	HGBM		0.005648
Table 3	A_81 RMSE	RF	RF		3.22e-06
Table 4	phi_162 MAE	DT	ERT		0.012507
Table 5	phi_162 RMSE	ERT	ERT		0.078591
Table 3	A_240 RMSE	ERT	RF		1.27e-05
Table 4	phi_240 MAE	DT	ERT		0.039941
Table 5	phi_240 RMSE	ERT	ERT		0.177287

The practical reading is blunt

- low-order gaps at 1 and 3 are now numerically tiny on amplitude, but they are still not closed;
- harmonic 0 remains structurally wrong on amplitude because the repository best family is still HGBM instead of the paper SVM branch;
- the major blockers remain phase-side gaps at 3, 162, and 240 ;
- 240 improved only on harmonic-family alignment, not on numeric closure.

Interpretation By Run Family

1. The Refresh Run Confirmed The Baseline

exact_open_cell_refresh_full_bank reproduced the current exact-paper baseline exactly:

- 5 Table 3 cells met
- 5 Table 4 cells met
- 4 Table 5 cells met
- 0 fully closed harmonics
- 7 partial harmonics
- 3 open harmonics

This means the campaign did not expose hidden regressions in the canonical full-bank branch.

2. The Two Best Runs Improved Only Harmonic Partiality

`exact_open_cell_paper_family_reference` and `exact_open_cell_phase_gap_bridge` both improved Table 6 status by making harmonic 240 partial instead of fully open.

But neither run

- closed a new numeric cell;
- reduced any open numeric gap below paper target;
- produced a first fully closed harmonic.

3. Two Repair Subsets Regressed Existing Good Cells

`exact_open_cell_low_order_repair` and `exact_open_cell_high_order_tree_repair` each regressed 3 previously met cells versus the canonical baseline.

This matters because it shows that family narrowing by itself is not enough. The exact-paper branch is sensitive to subset composition, and some supposedly targeted subsets destroy already-good cells without repairing the intended open ones.

4. Export Stability Is No Longer The Limiting Factor

All 6 runs completed with

- 0 failed ONNX exports
- 0 matched reference relative paths
- no surrogate-export requirement in this campaign

So the branch is operationally stable. The blocker is no longer exportability. The blocker is now real paper-target closure.

Main Conclusions

The campaign supports five conclusions.

1. The New RCIM Model-Bank Reproduction Framing Worked

The campaign was readable in the correct format:

- which cells stayed open;
- which harmonics stayed open or partial;
- which subsets regressed the baseline.

This is materially better than a winner-only campaign story.

2. The Campaign Did Not Solve Any Numeric Paper Gap

No Table 3-5 cell changed from `above_paper_target` to `met_paper_target`.

So, in the strict sense that matters for RCIM Model-Bank Reproduction, the campaign did not close any table cell.

3. Harmonic 240 Improved Only One Level

The best runs moved 240 from fully open to partial. That is real progress, but still insufficient because all three numeric cells at 240 remain above target.

4. The Main Blockers Are Still Structural

The unresolved branch is now clear

- `A_0` still wants the `SVM`-side paper behavior;
- `1` and `3` are very close numerically but still open;
- `162` and `240` remain the major phase-side blockers.

5. Future Work Must Be More Surgical Than Family Subsets Alone

This campaign shows that family-subset selection alone is too blunt. If the next iteration is meant to close real cells, it likely needs a more target-aware exact-paper workflow than the current family-only runner surface.

Recommended Next Actions

The next technically justified steps are

1. keep `exact_open_cell_paper_family_reference` as the bookkeeping winner for this campaign only;
2. keep the canonical `RCIM Model-Bank Reproduction` reading anchored to the still-open numeric cells listed above;
3. prioritize a next exact-paper iteration that can act more surgically on:
`A_0`, `phi_3`, `phi_162`, and the full `240` triplet;
4. treat `1` and `3` amplitude as near-threshold cells that may benefit from a target-aware refinement instead of another coarse family subset;
5. avoid reusing the low-order-only and high-order-tree-only subsets as-is, because both regressed already-good cells.

Artifact References

- Campaign root:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/`
- Campaign leaderboard:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/campaign_leaderboard.yaml`
- Campaign best run YAML:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/campaign_best_run.yaml`
- Campaign best run note:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/campaign_best_run.md`
- Winning validation summary:
`output/validation_checks/paper_reimplementation_rcim_exact_model_bank/forward/open_cell_repair/shared/2026-04-13-22-08-40__exact_open_cell_paper_family_reference_campaign_run/validation_summary.yaml`
- Winning model bundle:
`output/validation_checks/paper_reimplementation_rcim_exact_model_bank/forward/open_cell_repair/shared/2026-04-13-22-08-40__exact_open_cell_paper_family_reference_campaign_run/paper_family_model_bank.pkl`
- Per-run launcher logs:
`output/training_campaigns/track1/exact_paper/forward/uncategorized/shared/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/logs/`